

DHRUV MOTAVAL

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[GitHub](#) · [Hugging Face](#) · [LinkedIn](#) · [Portfolio](#)

TECHNICAL SKILLS

Languages: Python (primary), SQL, JavaScript

LLM & GenAI: LangChain, LangGraph, LCEL Pipelines, RAG Systems, Agents & Tool Calling, LangSmith (Tracing & Observability), Prompt & System-Prompt Design, OpenAI API, Groq API, Ollama

Retrieval & Vector: ChromaDB, Sentence Transformers, Embeddings, Semantic Search, Document Chunking, Vector Stores

Backend & Deployment: FastAPI, REST APIs, SQLite, PostgreSQL, Docker, Git, Linux

ML & Data: Scikit-learn, XGBoost, Pandas, NumPy

Speech & NLP: Whisper, spaCy

Interfaces: Streamlit, Gradio, Chatbot Development

EXPERIENCE

Web Developer Internship | iWebwala

Jan 2026 – Mar 2026

- Championed the full cycle development and launch of a solar installation company's website, including domain registration and Netlify deployment, leading to a 100% increase in online presence.
- Spearheaded development of an AI voice assistant that automated customer interactions for restaurant reservations and takeout orders, reducing average call duration by 3 minutes.
 - Launched a conversational AI pipeline on Railway, integrating Twilio Voice, Whisper, and ElevenLabs to automate 50% of initial customer inquiries..

PROJECTS

RAG Document Q&A System [Live Demo](#)

FastAPI · ChromaDB · Sentence Transformers · Groq · LangChain · LangSmith · Streamlit

- Architected a document Q&A pipeline end-to-end: PDF extraction, chunking, Sentence Transformer embeddings, ChromaDB retrieval, and answer generation via Groq's Llama-3.3-70B, served through a FastAPI backend with a Streamlit UI.
- Introduced a similarity-distance threshold so low-relevance queries return "not found in document" instead of a guessed answer.
- Logged every query to SQLite to track volume, unanswered questions, and latency: 1047ms average, 3087ms P95 across 50+ test queries.
- Implemented LangSmith tracing to pinpoint slow or low-quality pipeline steps, identifying and fixing a bottleneck in the retrieval step.

Multilingual Call Classifier [GitHub](#)

Whisper · Groq · spaCy · LangGraph · LangChain · SQLite

- Architected a LangGraph StateGraph replacing a linear script, integrating Whisper for transcription and language detection, routing 15% of international calls to expedited translation services.
- Extracted names and urgency keywords with spaCy, then classified each call as Open, Closed, or Urgent with a reasoned summary via structured LLM output.
- Replaced hardcoded if/else logic with a tool-calling action agent that reads the classification and transcript, selects from three tools (create_urgent_ticket, schedule_followup, close_ticket), and loops agent → tools → agent via tools_condition until done, logging every call to a SQLite ticket database.
- Wrote 6 tests mocking Whisper, spaCy, and the Groq client at the boundary so the graph's routing and tool-calling logic run offline in under a second, no API key or GPU required.

Stroke Risk Prediction API [Live Demo](#)

FastAPI · XGBoost · Scikit-learn · Pandas · Docker

- Trained and compared four models (Logistic Regression, Random Forest, Gradient Boosting, XGBoost) on 5,110 patient records, then shipped the best performer, XGBoost, at ROC-AUC of 0.813.
- Packaged it as a Dockerized FastAPI service that returns a high/low risk classification and stays live, covering the full path from trained model to running container.

ADDITIONAL PROJECTS

- LangChain Multi-Persona Chatbot** ([GitHub](#) · [Live Demo](#)): Built a streaming LangChain LCEL chatbot with 4 built-in personas plus a custom-prompt mode, switchable between two Groq LLaMA models (3.1 8B / 3.3 70B), and a 10-message sliding-window memory; **deployed on Hugging Face Spaces**.
- AI Meeting Minutes Generator** ([GitHub](#) · [Live Demo](#)): Constructed a Gradio application transcribing audio with Whisper, identifying attendees via spaCy, and outputting formatted PDFs, reducing manual effort by 90%.

EDUCATION

Parul Institute of Technology, Vadodara · 2023 – 2026

B.Tech, Artificial Intelligence · CGPA 7.12 / 10

Publication: "Limits of AI in Software Development: Empirical Analysis of AI vs Human Engineers," ISJEM Journal, April 2026.

A. Y. DadaBhai Institute of Technology, Kosamba · 2019 – 2023

Diploma, Computer Engineering · CGPA 7.91 / 10